Appendix

Additional Result

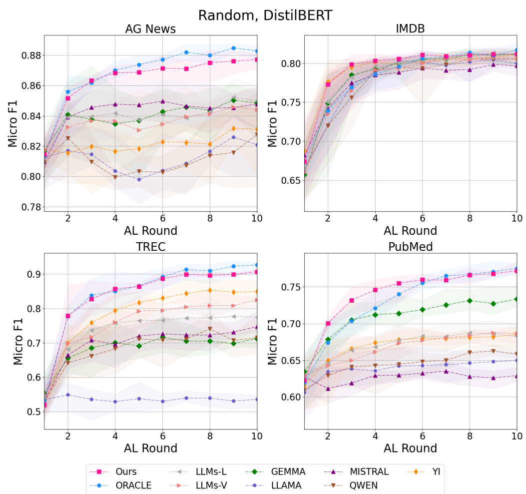

Figure 9: Averaged micro-F1 score with random on DistilBERT, averaged results with 5 random seeds.

Parameter Sensitive Analysis

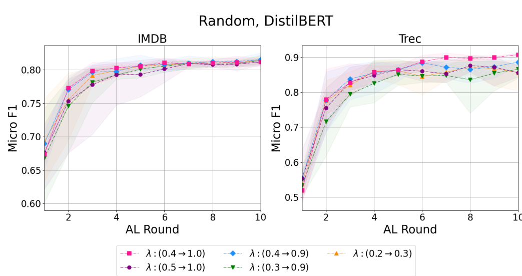

Figure 10: Performance on different negative learning weight parameter λ.

Figures 10 and 11 present the parameter sensitivity analysis for the negative learning weight λ and the annotation discrepancy weight α. The results are reported as micro-F1 scores, averaged over five random seeds using the Random query strategy on DistilBERT. Since λ is linearly increased during AL iterations, the legends indicate its starting and ending values. Overall, the results show that increasing λ benefits the learning process by gradually incorporating negative labels. Additionally, setting $\alpha = 0.5$ allows the model to effectively leverage annotation discrepancies, helping it focus on more reliable supervision.

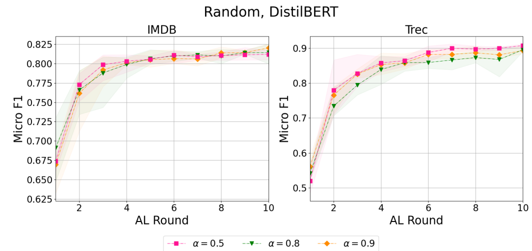

Figure 11: Performance with different values of the annotation discrepancy weight parameter α.

Qualitative Evaluation of MoLAM

```
"Label list":       ["negative", "positive"]
"Input text":       "excellent episode movie ala pulp fiction. 7 days -
                     7 suicides. it doesnt get more depressing than this.
                     movie rating : 8 \/ 10 music rating : 10 \/ 10"}
"GT label":         "positive"

"GEMMA-logits":     [0.9767777957,0.0229839049]
"GEMMA-consis.":    [1.0,0.0]
"LLAMA-logits":     [0.9156599855,0.085212484]
"LLAMA-consis.":    [1.0,0.0]
"MISTRAL-logits":   [0.9987564624,0.0000109721]
"MISTRAL-consis.":  [1.0,0.0]
"QWEN-logits":      [0.4611049891,0.5277478695]
"QWEN-consis.":     [0.5,0.5]
"YI-logits":        [0.0007171016,0.8430164922]
"YI-consis.":       [0.0,0.9]

"LLMs-L label":     "negative"
"LLMs-V label":     "negative"
"MoLAM label":      "positive"
```

(a) An example on IMDB.

```
"Label list":       ["world", "sports", "business", "science technology"]
"Input text":       "new mexico governor backs uc as los alamos lab manager
                    los angeles new mexico governor bill richardson today
                    endorsed the university of california to keep managing
                    the los alamos national laboratory."}
"GT label":         "science technology"

"GEMMA-logits":     [0.2373220176,0.0002307892,0.4434023127,0.3047787733]
"GEMMA-consis.":    [0.4,0.0,0.5,0.1]
"LLAMA-logits":     [0.2119691484,0.0002227128,0.5079409555,0.2716743574]
"LLAMA-consis.":    [0.1,0.0,0.7,0.2]
"MISTRAL-logits":   [0.0000087362,0.0000826602,0.0010006733,0.7540609537]
"MISTRAL-consis.":  [0.0,0.0,0.0,0.7]
"QWEN-logits":      [0.0050726533,0.0001707077,0.8523577452,0.1314268112]
"QWEN-consis.":     [0.0,0.0,1.0,0.0]
"YI-logits":        [0.0001573563,0.0000079763,0.000561323,0.9883978786]
"YI-consis.":       [0.0,0.0,0.0,1.0]

"LLMs-L label":     "science technology"
"LLMs-V label":     "business"
"MoLAM label":      "science technology"
```

(b) An example on AG News.

Figure 12: Qualitative comparison of annotation performance between LLM-ensemble and MoLAM.

Fig. 12 presents two examples illustrating the annotation performance of MoLAM. In these examples, LLMs-L and LLMs-V represent the LLM-ensemble baselines based on logits aggregation and majority voting, respectively. The values shown in square brackets correspond to the logits (for LLMs-L) or consistency scores (for LLMs-V) associated with each label index in the label list. In some cases, the sum of the consistency scores does not equal 1 because certain LLMs may generate non-existent or invalid labels.

All logits and consistency scores from the participating LLMs serve as input features to MoLAM. Notably, in both examples, MoLAM successfully predicts the correct label even when both ensemble baselines fail. This highlights the effectiveness of our proposed Mixture-of-LLMs-based annotation model in capturing nuanced decision patterns beyond simple aggregation, leading to more accurate and robust annotations.

CUDA Memory Usage

	AG News	IMDB	TREC	PubMed
DistilBERT	20489	20780	20348	20114
DistilRoBERTa	20516	20604	19988	20340

Table 5: The maximum CUDA memory occupation (MB) during the AL iteration across different query strategies.

MoLAM parameter

	AG News	IMDB	TREC	PubMed
Learning rate (lr)	0.07	0.01	0.05	0.01
Max depth (md)	5	5	6	3
# Estimators (ne)	300	300	300	500

Table 6: XGBoost hyperparameters used in MoLAM across datasets.

Prompt design

Classify the given question based on the following categories: **{List of labels}**
Task: Determine the most appropriate category for the question. Your response should be only one of these labels: **{List of labels}**, with no additional text or explanation.
Question: **{article}**
Output: